

# EXPERIMENT - 13

## 1. Problem Statement

Write a menu-driven C program using `switch-case` to input a positive integer and perform one of the following operations based on the user's choice:

- **Option 1** — Count the number of even digits and odd digits.  
*E.g.* Number: 12045 → Even\_dig = 3, Odd\_dig = 2
- **Option 2** — Find the sum of its even digits and the sum of its odd digits separately.  
*E.g.* Number: 12345 → Sum\_even = 6, Sum\_odd = 9

## 2. Logic Used

Both operations need the digits of the number one by one. Two statements do all the work:

| Statement                       | Meaning                                          |
|---------------------------------|--------------------------------------------------|
| <code>digit = temp % 10;</code> | Gives the <b>last digit</b> of the number        |
| <code>temp = temp / 10;</code>  | <b>Removes</b> the last digit (integer division) |

The loop repeats until `temp` becomes 0. For 12045 the digits are extracted in the order **5 → 4 → 0 → 2 → 1**.

A digit is **even** when `digit % 2 == 0`, otherwise it is **odd**. Note that **0 is an even digit**, which is why 12045 has 3 even digits (2, 0, 4).

## 3. Algorithm

```
Step 1 : START

Step 2 : Display the menu
        1. Count even and odd digits
        2. Sum of even digits and sum of odd digits

Step 3 : Read choice

Step 4 : Read num

Step 5 : If num <= 0 then
        Print "Error: not a positive integer"
        Go to Step 10
    End If

Step 6 : switch (choice)

Step 7 : case 1:                                [ COUNT ]
        Set even_dig = 0, odd_dig = 0, temp = num
        While temp > 0 repeat
            digit = temp % 10
```

```
        If digit % 2 == 0 then
            even_dig = even_dig + 1
        Else
            odd_dig = odd_dig + 1
        End If
        temp = temp / 10
    End While
    Print even_dig, odd_dig
    break
```

```
Step 8 : case 2: [ SUM ]
        Set sum_even = 0, sum_odd = 0, temp = num
        While temp > 0 repeat
            digit = temp % 10
            If digit % 2 == 0 then
                sum_even = sum_even + digit
            Else
                sum_odd = sum_odd + digit
            End If
            temp = temp / 10
        End While
        Print sum_even, sum_odd
        break
```

```
Step 9 : default:
        Print "Invalid choice"
        break
```

```
Step 10 : STOP
```

## 4. Test Case Table

| TC # | Choice | Input Number | Expected Output             | Actual Output                 | Result |
|------|--------|--------------|-----------------------------|-------------------------------|--------|
| TC1  | 1      | 12045        | Even_dig = 3<br>Odd_dig = 2 | Even_dig = 3<br>Odd_dig = 2   | PASS   |
| TC2  | 2      | 12345        | Sum_even = 6<br>Sum_odd = 9 | Sum_even = 6<br>Sum_odd = 9   | PASS   |
| TC3  | 1      | 1000         | Even_dig = 3<br>Odd_dig = 1 | Even_dig = 3<br>Odd_dig = 1   | PASS   |
| TC4  | 3      | 123          | Invalid choice message      | Invalid choice! Enter 1 or 2. | PASS   |

### Why these test cases were chosen

| TC # | Purpose of the test case                                                             |
|------|--------------------------------------------------------------------------------------|
| TC1  | First example given in the question — normal case for menu option 1 (counting)       |
| TC2  | Second example given in the question — normal case for menu option 2 (summing)       |
| TC3  | Number containing zeros — checks that 0 is correctly counted as an <b>even</b> digit |
| TC4  | Invalid menu choice — tests the <code>default:</code> block of the switch statement  |

## 5. Source Code

**File name:** `digit_operations.c` (71 lines)

```
#include <stdio.h>

int main()
{
    int choice, digit;
    long num, temp;
    int even_dig, odd_dig;
    int sum_even, sum_odd;

    printf("===== MENU =====\n");
    printf("1. Count even and odd digits\n");
    printf("2. Sum of even digits and sum of odd digits\n");
    printf("===== \n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("Error: %ld is not a positive integer.\n", num);
        return 1;
    }

    switch (choice) {

        case 1:
            even_dig = 0;
            odd_dig = 0;
            temp = num;

            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_dig++;
                else
                    odd_dig++;
                temp = temp / 10;
            }

            printf("\nNumber : %ld\n", num);
            printf("Even_dig = %d\n", even_dig);
            printf("Odd_dig = %d\n", odd_dig);
            break;

        case 2:
            sum_even = 0;
            sum_odd = 0;
            temp = num;

            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 == 0)
```

```
        sum_even = sum_even + digit;
    else
        sum_odd = sum_odd + digit;
    temp = temp / 10;
}

printf("\nNumber : %ld\n", num);
printf("Sum_even = %d\n", sum_even);
printf("Sum_odd = %d\n", sum_odd);
break;

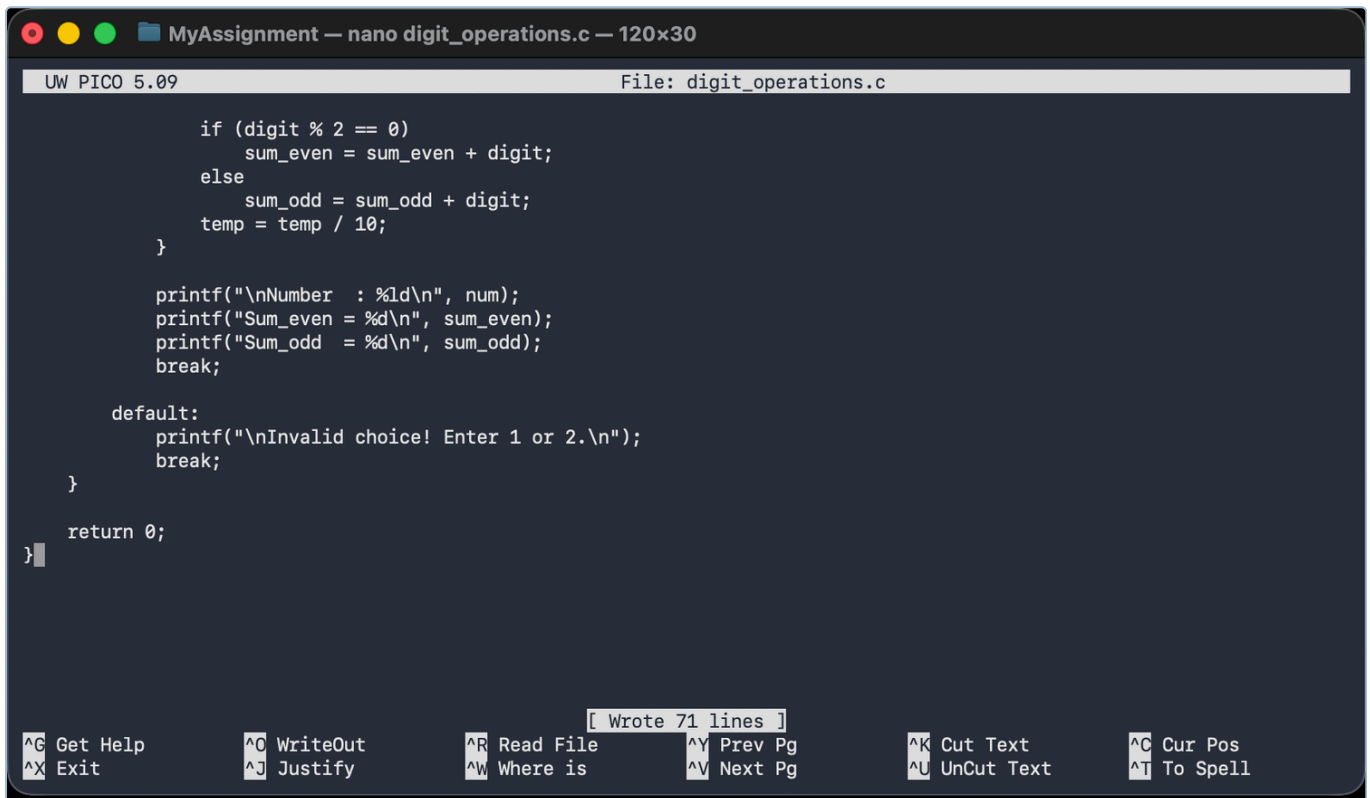
default:
    printf("\nInvalid choice! Enter 1 or 2.\n");
    break;
}

return 0;
}
```

## 6. Writing the Program in the Terminal

The program was written directly in the macOS Terminal using the `nano` editor:

```
$ cd ~/Desktop
$ mkdir MyAssignment
$ cd MyAssignment
$ nano digit_operations.c
```



```
UW PICO 5.09                                File: digit_operations.c

        if (digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;
        temp = temp / 10;
    }

    printf("\nNumber : %ld\n", num);
    printf("Sum_even = %d\n", sum_even);
    printf("Sum_odd = %d\n", sum_odd);
    break;

default:
    printf("\nInvalid choice! Enter 1 or 2.\n");
    break;
}

return 0;
}
```

[ Wrote 71 lines ]

|                    |                    |                     |                   |                      |                    |
|--------------------|--------------------|---------------------|-------------------|----------------------|--------------------|
| <b>^G</b> Get Help | <b>^O</b> WriteOut | <b>^R</b> Read File | <b>^Y</b> Prev Pg | <b>^K</b> Cut Text   | <b>^C</b> Cur Pos  |
| <b>^X</b> Exit     | <b>^J</b> Justify  | <b>^W</b> Where is  | <b>^V</b> Next Pg | <b>^U</b> UnCut Text | <b>^T</b> To Spell |

## 7. Source Code Listing in the Terminal

```
MyAssignment — -zsh — 120x30
adilshakil@ADILs-MacBook-Air MyAssignment % cd ~/Desktop
adilshakil@ADILs-MacBook-Air MyAssignment % head -25 digit_operations.c
#include <stdio.h>

int main()
{
    int choice, digit;
    long num, temp;
    int even_dig, odd_dig;
    int sum_even, sum_odd;

    printf("===== MENU =====\n");
    printf("1. Count even and odd digits\n");
    printf("2. Sum of even digits and sum of odd digits\n");
    printf("===== \n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("Error: %ld is not a positive integer.\n", num);
        return 1;
    }

    switch (choice) {
adilshakil@ADILs-MacBook-Air MyAssignment %
```

```
MyAssignment — -zsh — 120x30
adilshakil@ADILs-MacBook-Air MyAssignment % sed -n 26,50p digit_operations.c

        case 1:
            even_dig = 0;
            odd_dig = 0;
            temp = num;

            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_dig++;
                else
                    odd_dig++;
                temp = temp / 10;
            }

            printf("\nNumber : %ld\n", num);
            printf("Even_dig = %d\n", even_dig);
            printf("Odd_dig = %d\n", odd_dig);
            break;

        case 2:
            sum_even = 0;
            sum_odd = 0;
            temp = num;

adilshakil@ADILs-MacBook-Air MyAssignment %
```

```
MyAssignment — -zsh — 120x30
[adilshakil@ADILs-MacBook-Air MyAssignment % tail -21 digit_operations.c
    while (temp > 0) {
        digit = temp % 10;
        if (digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;
        temp = temp / 10;
    }

    printf("\nNumber : %ld\n", num);
    printf("Sum_even = %d\n", sum_even);
    printf("Sum_odd = %d\n", sum_odd);
    break;

default:
    printf("\nInvalid choice! Enter 1 or 2.\n");
    break;
}

return 0;
}
[adilshakil@ADILs-MacBook-Air MyAssignment % ]
```

## 8. Compilation, Execution & Output

The program was compiled with the GCC compiler and executed in the Terminal:

```
$ gcc digit_operations.c -o digit_operations
$ ./digit_operations
```

The program compiled successfully with **no errors and no warnings**.

```
MyAssignment — -zsh — 120x30

[adilshakil@ADILs-MacBook-Air Desktop % cd ~/Desktop
[adilshakil@ADILs-MacBook-Air Desktop % mkdir MyAssignment
[adilshakil@ADILs-MacBook-Air Desktop % cd MyAssignment
[adilshakil@ADILs-MacBook-Air MyAssignment % nano digit_operations.c
[adilshakil@ADILs-MacBook-Air MyAssignment % gcc digit_operations.c -o digit_operations
[adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_operations
===== MENU =====
1. Count even and odd digits
2. Sum of even digits and sum of odd digits
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 12045

Number : 12045
Even_dig = 3
Odd_dig = 2
[adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_operations
===== MENU =====
1. Count even and odd digits
2. Sum of even digits and sum of odd digits
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 12345

Number : 12345
Sum_even = 6
Sum_odd = 9
adilshakil@ADILs-MacBook-Air MyAssignment %
```

```
MyAssignment — -zsh — 120x30

[adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_operations
===== MENU =====
1. Count even and odd digits
2. Sum of even digits and sum of odd digits
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 1000

Number : 1000
Even_dig = 3
Odd_dig = 1
[adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_operations
===== MENU =====
1. Count even and odd digits
2. Sum of even digits and sum of odd digits
=====
Enter your choice (1 or 2): 3
Enter a positive integer: 123

Invalid choice! Enter 1 or 2.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

## 9. Conclusion

The menu-driven program was successfully written, compiled and executed on macOS using the GCC compiler. All **4 test cases passed**, including the case of a number containing zeros (which verifies that 0 is treated as an even digit) and an invalid menu choice (which verifies the `default` block of the switch). The use of `switch-case` allows the required operation to be

selected cleanly at run time, and the `% 10` / `/ 10` technique correctly extracts every digit of the entered number.